

June 19, 2026

Admissions Committee
Norwegian University of Science and Technology (NTNU)
Ålesund, Norway

Dear Admissions Committee,

I am writing to express my profound interest in the PhD Candidate position focusing on Secure and Trustworthy Data Sharing for Maritime AI Model Development. With a strong background in AI, data engineering, and the mathematical modeling of physical systems, I am highly motivated to tackle the cybersecurity and data-sovereignty bottlenecks preventing the global deployment of Maritime AI.

In my State Engineer thesis ("Gas Turbine Digital Twin"), I engineered predictive maintenance architectures to forecast the degradation of complex industrial machinery. A major challenge in that work was handling highly sensitive telemetry that industrial partners were reluctant to share. To solve similar issues in my Master's thesis ("Cardiovascular Medical Digital Twin"), I developed architectures (Medical_MambaTS and PI-DeepONet) that mathematically enforce physical laws directly into the neural network.

I propose utilizing this background to build a secure, Federated Learning (FL) framework for the Norwegian Maritime AI Center. By training models directly on the edge (aboard ships) and only sharing encrypted parameter updates, we can guarantee data sovereignty. Furthermore, by embedding the physics of maritime vessels (e.g., hydrodynamics) directly into the AI loss function, we can automatically detect and reject cyber-attacks and adversarial data poisoning, as malicious inputs will fundamentally violate physical reality.

I am highly proficient in Python, experienced with complex distributed systems, and eager to collaborate with your industrial partners in Ålesund.

Thank you for your consideration.

Sincerely,
Ismail Aissaoui
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